

The Intention Experiment (full text) by Lynne McTaggart. In HTML for free access. Part 1 of 4.

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This is a complete reprint of the book titled “The Intention Experiment” by Lynne McTaggart. It is a non-fiction book, and it is groundbreaking. In this book, the author has compiled all those studies about the reality of ESP, and PSI, and compiled the results. The results are pretty damning. Something is going on, and Newtonian physics cannot explain it. It can only be explained with quantum physics.

What is going on is that quantum physics is working and weaving it’s magic throughout our lives, and rather than discount things as “superstition” and out-dated religion, this book connects actual scientific studies with the quantum physics principles involved. It explains so many thing that have been discounted as pure superstition.

Thus it’s placement in my blog.

This is for those people who want nice and clean answers to what is going on, yet cannot shake off the Newtonian physics that they learned in High School. This book teaches you that there is a deeper reality behind everything and as such, it helps

explain some elements of paranormal and religion that are often discounted as primitive nonsense.

Welcome to the world of quantum physics and how all those things about prayer, intention, and spirituality actually does have a scientific foundation that they are based upon.

Some Comments

One of the best books I have ever read. You will learn so much about intention. I call it "desire". How important this is!

-[Carol S. Burney](#)

I really enjoyed this book. I was impressed with the author's ability to make complex science clear and also the use of credible sources.

She really doesn't talk so much about using your thoughts to change your life. It is more of a book about how science, real science, is showing more and more how the human mind is seen by quantum physics and other legitimate scientific disciplines.

It was really interesting to me to see that the mind really is much more powerful than we think it is.

As I said, I was very impressed with the research she used in the book as the references were to legitimate experiments that had been peer reviewed. Good insights into the amazing power of intention.

-Zipporia

I rate this a solid 5 due to the importance of the message; it combines everything I have learned in pieces into a nice little package.

Although not too much attention was paid to the way the writing flows, and as some of you have pointed out the sloppiness of the writing, I have to say that I also write sloppy when I discover a very cool thing.

The excitement overwhelms my style and attention to detail. But otherwise, it is a very solid read and I did not read it to be blown away with the "literature" quality of the writing, but for the clear and to the point message that it communicates so clearly.

This is a life altering book if it's the first book you are reading about the power of your intentions, thoughts and quantum physics. Awesome job!

-Netaron

In Quantum theory , "the world" is comprised of two "systems",the system containing the observer and the system containing what is observed.

Until the observer focuses his attention on the observed system it exists only as a host of infinite possibilities. The observer observation or measurement "fixes" its reality. That is the scientific theory.

Hard enough for us mere mortals to grasp or integrate it with what we have been taught.

From quantum theory it is almost irresistible to move to a consideration of how intention, rather than mere observation or objective measurement might work in our world and that is what this thought provoking book does.

The experiments are about whether intention can change outcome or even in one experiment, can change a previously measured reality.

It explores the power of consciousness and invites readers, through excercises in the book and through the web site to become part of an ongoing living experiment in consciousness.

I found the experiments fascinating, engaging and worthy of reflection..and I found reflection on them energizing. I recommend it enthusiastically for anyone who is at work on increasing their own awareness and trying to live fully present in the moment.

[-Lindsay N. Bowker](#)

For those interested in Quantum Science, in the Zero Point Field, and in what they call the "soup of creation" - this is a "must read".

I was pleased with this book right up to the chapter called Praying for Yesterday, which introduced experiments that couldn't hold water logically. I was so frustrated by that chapter that I almost ditched the book entirely - BUT - the third paragraph from the end of that chapter is a prize so wondrous - a double concept so unbelievable and empowering that all else is forgiven.

In fact, based solely on that, I ordered her first book - The Field.

If you're one of the lucky one's who can wrap your brain around concepts like these - ordering this book is a favor you need to do for yourself. It's an exciting rush of potential at your command - if you allow it to be so. I, personally, noting the flaws of the above chapter, would still recommend this book without hesitation as a "must read"!

[-SciFiCahill](#)

Our thoughts create our reality.

This is a well written book about quantum physics and its extraordinary implications. That we live in a "Field" (Zero Point Field) which is a constant dance of quantum energy exchange. Clearly we are connected to the entire universe through our pulsating energy which is constantly interacting with the vast energy "out there".

This is a great book filled with information about how the universe operates and our connection to it. This vast flow of energy, Consciousness, if you will, is all around us and is our connection to the "Source" or "Creator" of the Universe. A very powerful book, which I highly recommend.

-Richard Grant

The Intention Experiment

The preface is in italicized purple font. It's pretty boring, but it tells you how the book came about, and at that it can give you some insight in how some things can manifest in our universe. Don't ever think that everything is a coincidence.

Preface

THIS BOOK REPRESENTS A PIECE of unfinished business that began 2001 when I published a book called The Field. In the course of trying to find a scientific explanation for homeopathy and spiritual healing, I had inadvertently uncovered the makings of a new science.

During my research, I stumbled across a band of frontier scientists who had spent many years re-examining quantum physics and its extraordinary implications. Some had resurrected certain equations regarded as superfluous in standard quantum physics. These equations, which stood for the Zero Point Field, concerned the extraordinary quantum field generated by the endless passing back and forth of energy between all subatomic particles. The existence of the Field implies that all matter in the universe is connected on the subatomic level through a constant dance of quantum energy exchange.

Other evidence demonstrated that, on the most basic level, each one of us is also a packet of pulsating energy constantly interacting with this vast energy sea.

But the most heretical evidence of all concerned the role of consciousness. The well-designed experiments conducted by these scientists suggested that consciousness is a substance outside the confines of our bodies – a highly ordered energy with the capacity to change physical matter. Directing thoughts at a target seemed capable of altering machines, cells and, indeed, entire multicelled organisms like human beings. This mind-over-matter power even seemed to traverse time and space.

In The Field I aimed to make sense of all the ideas resulting from these disparate experiments and to synthesize them into one generalized theory. The Field created a picture of an interconnected universe and a scientific explanation for many of the

most profound human mysteries, from alternative medicine and spiritual healing to extrasensory perception and the collective unconscious.

The Field apparently hit a nerve. I received hundreds of letters from readers who told me that the book had changed their lives. A writer wanted to depict me as a character in her novel. Two composers wrote musical compositions inspired by it, one of which was played on the international stage.

I was featured in a movie, What the Bleep!?! Down the Rabbit Hole, and on the What The Bleep Do We Know!?! Calendar, released by the film's producers. Quotes from The Field became the centrepiece of a printed Christmas card.

However gratifying this reaction, I felt that my own journey of discovery had hardly left the station platform. The scientific evidence I had amassed for The Field suggested something extraordinary and even disturbing: directed thought had some sort of central participatory role in creating reality.

Targeting your thoughts – or what scientists ponderously refer to as ‘intention’ and ‘intentionality’ – appeared to produce an energy potent enough to change physical reality. A simple thought seemed to have the power to change our world.

After writing The Field, I puzzled over the extent of this power and the numerous questions it raised. How, for instance, could I translate what had been confirmed in the laboratory for use in the world that I lived in? Could I stand in the middle of a railway track and, Superman-style, stop the 9:45 to Paddington with my thoughts? Could I fly myself up to fix my roof with a bit of directed thought? Would it now be possible to cross doctors and healers off my list of essential contacts, seeing as I might now be able to think myself well? Could I help my children pass their maths tests just by thinking about it? If linear time and three-dimensional space didn't really exist, could I go back and erase all those moments in my life that had left me with lasting regret? And could my one puny bit of mental input do anything to change the vast catalogue of suffering on the planet?

The implications of this evidence were unsettling. Should we be minding every last thought at every moment? Was a pessimist's view of the world likely to be a self-fulfilling prophecy? Were all those negative thoughts – that ongoing inner dialogue of judgement and criticism – having any effect outside our heads?

Were there conditions that improved your chances of having a better effect with your thoughts? Would a thought work any old time or would you, your intended target and indeed the universe itself have to be in the mood? If everything is affecting everything else at every moment, doesn't that counteract and thereby nullify any real effect?

What happens when a number of people think the same thought at the same time? Would that have an even larger effect than thoughts generated singly? Was there a threshold size that a group of like-minded intenders had to reach in order to exert the most powerful effect? Was an intention 'dose dependent' – the larger the group, the larger the effect?

*An enormous body of literature, starting with *Think and Grow Rich* [1] by Napoleon Hill, arguably the first self-actualization guru, has been generated about the power of thought. 'Intention' has become the latest New Age buzzword. Practitioners of alternative medicine speak of helping patients heal 'with intention'. Even Jane Fonda writes about raising children 'with intention'.[2]*

What on earth, I wondered, was meant by 'intention'? And how exactly can one become an efficient 'intender'? The bulk of the popular material had been written off the cuff – a smattering of Eastern philosophy here, a soupçon of Dale Carnegie there, with very little scientific evidence that it worked.

*To find answers to all of these questions, I turned, once again, to science, scouring the scientific literature for studies on distant healing or other forms of psychokinesis, or mind over matter. I sought out international scientists who experimented with how thoughts can affect matter. The science described in *The Field* had been carried out mainly in the 1970s; I examined more recent discoveries in quantum physics for further clues.*

I also turned to those people who had managed to master intention and who could perform the extraordinary – spiritual healers, Buddhist monks, Qigong masters, shamans – in order to understand the transformational processes they underwent to be able to use their thoughts to powerful effect. I uncovered myriad ways that intention is used in real life – in sports, for instance, and during healing modalities such as biofeedback. I studied how native populations incorporated directed thought into their daily ritual.

I then began to dig up evidence that multiple minds trained on the same target magnified the effect produced by an individual. The evidence was tantalizing, mostly gathered by the Transcendental Meditation organization, suggesting that a group of likeminded thoughts created some sort of order in the otherwise random Zero Point Field.

At that point in my journey, I ran out of pavement. All that stretched before me, as far as I could tell, was uninhabited open terrain.

Then one evening, my husband Bryan, a natural entrepreneur in most situations, put forward what seemed to be a preposterous suggestion: 'Why don't you do some

group experiments yourself?’

I am not a physicist. I am not any kind of scientist. The last experiment I had conducted had been in a 10th grade science lab.

What I did have, though, was a resource available to few scientists: a potentially huge experimental body. Group intention experiments are extraordinarily difficult to perform in an ordinary laboratory. A researcher would need to recruit thousands of participants. How would he find them? Where would he put them? How would he get them all to think the same thing at the same time?

A book’s readers offer an ideal self-selected group of likeminded souls who might be willing to participate in testing out an idea. Indeed, I already had my own large population of regular readers with whom I communicated through e-news and my other spin-off activities from The Field.

I first broached the idea of carrying out my own experiment with dean emeritus of the Princeton University School of Engineering Robert Jahn and his colleague psychologist Brenda Dunne, who run the Princeton Engineering Anomalous Research (PEAR) laboratory, both of whom I had got to know through my research for The Field. Jahn and Dunne have spent some 30 years painstakingly amassing some of the most convincing evidence about the power of directed intention to affect machinery. They are absolute sticklers for scientific method, no-nonsense and to the point. Robert Jahn is one of the few people I have ever met who speaks in perfect, complete sentences. Brenda Dunne is equally perfectionist about detail in both experiment and language. I would be assured of no sloppy protocol in my experiments if Jahn and Dunne agreed to be involved.

The two of them also have a vast array of scientists at their disposal. They head the International Consciousness Research Laboratory, many of whose members are among the most prestigious scientists performing consciousness research in the world. Dunne also runs PEARTree, a group of young scientists interested in consciousness research.

Everyone met on occasions and kicked around some possibilities. Eventually, they put forward Fritz- Albert Popp, assistant director of the International Institute of Biophysics (IIB) in Neuss, Germany, to conduct the first intention experiments. I knew Fritz Popp through my research for The Field. He was the first to discover that all living things emit a tiny current of light. As a noted German physicist recognized internationally for his discoveries, Popp would also be a stickler for pristine scientific method.

Other scientists, such as psychologist Gary Schwartz of the Biofield Center at the University of Arizona, Marilyn Schlitz, vice president for research and education at the Institute of Noetic Sciences, Dean Radin, IONS' senior scientist, an psychologist Roger Nelson of the Global Consciousness Project, have also offered to participate.

I do not have any hidden sponsors of this project. The website and all our experiments will be funded by the proceeds of this book or grants, now and in the future.

Scientists involved in experimental research often cannot venture beyond their findings to consider the implications of what they have uncovered. Consequently, when assembling the evidence that already exists about intention, I have tried to consider the larger implications of this work and to synthesize these individual discoveries into a coherent theory. In order to describe in words concepts that are generally depicted through mathematical equations, I have had to reach for metaphoric approximations of the truth. At times, with the help of many of the scientists involved, I have also had to engage in speculation. It is important to recognize that the conclusions arrived at in this book represent the fruits of frontier science. These ideas are a work in progress. Undoubtedly new evidence will emerge to amplify and refine these initial conclusions.

Researching the work of people at the very forefront of scientific discovery again has been a humbling experience for me. Within the unremarkable confines of a laboratory, these largely unsung men and women engage in activities that are nothing short of heroic. They risk losing grants, academic posts and, indeed, entire careers groping alone in the dark. Most scratch around for grant money to enable them to carry on.

All advancements in science are somewhat heretical, each important new discovery partly, if not completely, negating the prevailing views of the day. To be a true explorer in science – to follow the unprejudiced lead of pure scientific inquiry – is to be unafraid to propose the unthinkable, and to prove friends, colleagues and scientific paradigms wrong. Hidden within the cautious, neutral language of experimental data and mathematical equation is nothing less than the makings of a new world, which slowly takes shape for all the rest of us, one painstaking experiment at a time.

Lynne McTaggart, June 2006

Notes – Preface

1. N. Hill, *Think and Grow Rich: The Andrew Carnegie Formula for Money Making*, New York: Ballantine Books (reissue edn), 1987.

Introduction

THE INTENTION EXPERIMENT is no ordinary book, and you are no ordinary reader. This is a book without an ending, for I intend for you to help me finish it. You are not only the audience of this book, but also one of its protagonists – the primary participants in cutting-edge scientific research. You, quite simply, are about to embark on the largest mind-over-matter experiment in history.

The Intention Experiment is the first ‘living’ book in three-dimensions. The book, in a sense, is a prelude, and the ‘contents’ carry on well beyond the time you finish the final page. In the book itself, you will discover scientific evidence about the power of your own thoughts, and you will then be able to extend beyond this information and test further possibilities through a massive, ongoing international group experiment, under the direction of some of the most well-respected international scientists in consciousness research. Through *The Intention Experiment*’s website (www.theintentionexperiment.com), you and the rest of the readers of this book will be able to participate in remote experiments, the results of which will be posted on the site. Each of you will become a scientist at the hub of some of the most daring consciousness experiments ever conducted.

The Intention Experiment rests on an outlandish premise: thought affects physical reality.

It’s not at all outlandish. Thought actually does create reality.

A sizeable body of research exploring the nature of consciousness, carried on for more than 30 years in prestigious scientific institutions around the world, shows that thoughts are capable of affecting everything from the simplest machines to the most complex living beings.[1]

This evidence suggests that human thoughts and intentions are an actual physical ‘something’ with the astonishing power to change our world. Every thought we have is a tangible energy with the power to transform.

A thought is not only a thing; a thought is a thing that influences other things.

If quanta were like fine particles of dust, or finely ground flour... then thoughts are like a breeze that attracts or scatters the dust particles everywhere.

This central idea, that consciousness affects matter, lies at the very heart of an irreconcilable difference between the world view offered by classical physics – the science of the big, visible world – and that of quantum physics – the science of the

world's most diminutive components. That difference concerns the very nature of matter and the ways it can be influenced to change.

The idea behind quantum physics is that consciousness and thoughts affect physical matter.

All of classical physics, and indeed the rest of science, is derived from the laws of motion and gravity developed by Isaac Newton in his *Principia*.

Newton's laws described a universe in which all objects moved within the three-dimensional space of geometry and time according to certain fixed laws of motion. Matter was considered inviolate and self-contained, with its own fixed boundaries. Influence of any sort required something physical to be done to something else – a force or collision. Making something change basically entailed heating it, burning it, freezing it, dropping it or giving it a good swift kick.

Newtonian laws, science's grand 'rules of the game', as the celebrated physicist

Richard Feynman once referred to them,[3] and their central premise, that things exist independently of each other, underpin our own philosophical view of the world. We believe that all of life and its tumultuous activity carries on around us, regardless of what we do or think. We sleep easy in our beds at night, in the certainty that when we close our eyes, the universe doesn't disappear.

Nevertheless, that tidy view of the universe as a collection of isolated, well-behaved objects got dashed in the early part of the twentieth century, once the pioneers of quantum physics began peering closer into the heart of matter. The tiniest bits of the universe, those very things that make up the big, objective world, did not in any way behave themselves according to any rules that these scientists had ever known.

This outlaw behavior was encapsulated in a collection of ideas that became known as the Copenhagen Interpretation, after the place where the forceful Danish physicist Niels Bohr and his brilliant protégé, the German physicist Werner Heisenberg, formulated the likely meaning of their extraordinary mathematical discoveries. Bohr and Heisenberg realized that atoms are not little solar systems of billiard balls but something far more messy: a tiny cloud of probability.

Every subatomic particle is not a solid and stable thing, but exists simply as a potential of any one of its future selves – or what is known by physicists as a 'superposition', or sum, of all probabilities, like a person staring at himself in a hall of mirrors.

One of their conclusions concerned the notion of 'indeterminacy'; that you can never know all there is to know about a subatomic particle all at the same time. If you

discover information about where it is, for instance, you cannot work out at the same time exactly where it is going or at what speed. They spoke about a quantum particle as both a particle – a congealed, set thing – and a ‘wave function’ – a big smeared-out region of space and time, any corner of which the particle may occupy. It was akin to describing a person as comprising the entire street where he lives.

Their conclusions suggested that, at its most elemental, physical matter isn’t solid and stable – indeed, isn’t an *anything* yet.

Subatomic reality did not resemble the solid and reliable state of being described to us by classical science, but an ephemeral prospect of seemingly infinite options. So capricious seemed the smallest bits of nature that the first quantum physicists had to make do with a crude symbolic approximation of the truth – a mathematical range of all possibility.

At the quantum level, reality resembled unset jelly.

Newtonian physics treated things as nice set fixed and solid objects; like billiard balls. That they would follow set rules of behavior. Quantum physics says otherwise. The smallest things are actually like unset jello. When you think about them, they turn hard and freeze in place.

The quantum theories developed by Bohr, Heisenberg and a host of others rocked the very foundation of the Newtonian view of matter as something discrete and self-contained. They suggested that matter, at its most fundamental, could not be divided into independently existing units and indeed could not even be fully described. Things had no meaning in isolation, but only in a web of dynamic interrelationship.

The quantum pioneers also discovered the astonishing ability of quantum particles to influence each other, despite the absence of all those usual things that physicists understand are responsible for influence, such as an exchange of force occurring at a finite velocity. Once in contact, particles retained an eerie remote hold over each other.

The actions – for instance, the magnetic orientation – of one subatomic particle instantaneously influenced the other, no matter how far they were separated.

In quantum physics, things influence other things regardless of physical distance. At the subatomic level, change also resulted through dynamic shifts of energy; these little packets of vibrating energy constantly traded energy back and forth to each other like ongoing passes in a game of basketball, a ceaseless to-ing and from-ing that gave rise to an unfathomably large basic layer of energy in the universe.[4]

Subatomic matter appeared to be involved in a continual exchange of information, causing constant refinement and subtle alteration. The universe was not a storehouse of static, separate objects, but a single organism of interconnected energy fields in a constant state of becoming. At its infinitesimal level, our world resembled a vast network of quantum information, with all its component parts constantly on the phone.

The only thing dissolving this little cloud of probability into something solid and measurable was the involvement of an observer.

Once these scientists decided to have a closer look at a subatomic particle by taking a measurement, the subatomic entity that existed as pure potential would ‘collapse’ into one particular state.

The implications of these early experimental findings were profound: living consciousness somehow was the influence that turned the possibility of something into something real. The moment we looked at an electron or took a measurement, *it appeared that we helped to determine its final state*. This suggested that the most essential ingredient in creating our universe is the consciousness that observes it. Several of the central figures in quantum physics argued that the universe was democratic and participatory – a joint effort between observer and observed. [5]

The observer effect in quantum experimentation gives rise to another heretical notion: that living consciousness is somehow central to this process of transforming the unconstructed quantum world into something resembling everyday reality. It suggests not only that the observer brings the observed into being, but also that nothing in the universe exists as an actual ‘thing’ independently of our perception of it.

Nothing in this universe exists without an observer to think about it.

It implies that observation – the very involvement of consciousness – gets the jelly to set.

It implies that reality is not fixed, but fluid, or mutable, and hence possibly open to influence.

The idea that consciousness creates and possibly even affects the physical universe also challenges our current scientific view of consciousness, which developed from the theories of the seventeenth-century philosopher René Descartes – mind is separate and somehow different from matter – and eventually embraced the notion that consciousness is entirely generated by the brain and remains locked up in the skull.

Most modern workaday physicists shrug their shoulders over this central conundrum: that big things are separate, but the tiny building blocks they are made up of are in

instant and ceaseless communication with each other. For half a century, physicists have accepted, as though it makes perfect sense, that an electron behaving one way subatomically somehow transmutes into 'classical' (that is, Newtonian) behavior once it realizes it is part of a larger whole.

In the main, scientists have stopped caring about the troublesome questions posed by quantum physics, and left unanswered by its earliest pioneers.

Quantum theory works mathematically. It offers a highly successful recipe for dealing with the subatomic world. It helped to build atomic bombs and lasers, and to deconstruct the nature of the sun's radiation. Today's physicists have forgotten about the observer effect.

They content themselves with their elegant equations and await the formulation of unified Theory of Everything or the discovery of a few more dimensions beyond the ones that ordinary humans perceive, which they hope will somehow pull together all these contradictory findings into one centralized theory.

Thirty years ago, while the rest of the scientific community carried on by rote, a small band of frontier scientists at prestigious universities around the globe paused to consider the metaphysical implications of the Copenhagen Interpretation and the observer effect.[6]

If matter was mutable, and consciousness *made* matter a set something, it seemed likely that consciousness might also be able to nudge things in a particular direction.

If you can control your thoughts, then you can control matter and the events in your life.

Their investigations boiled down to a simple question: if the act of *attention* affected physical matter, what was the effect of *intention* – of deliberately attempting to make a change? In our act of participation as an observer in the quantum world, we might be not only creators, but also influencers.[7](#)

They began designing and carrying out experiments, testing what they gave the unwieldy label of 'directed remote mental influence' or 'psychokinesis', or, in shorthand, 'intention' or even 'intentionality'.

A textbook definition of intention characterizes it as 'a purposeful plan to perform an action, which will lead to a desired outcome',[8] unlike a desire, which means simply focusing on an outcome, without a purposeful plan of how to achieve it.

Intention is quite different from desire.

An intention was directed at the intender's own actions; it required some sort of reasoning; it required a commitment to do the intended deed. Intention implied

purposefulness: an understanding of a plan of action and a planned satisfactory result.

Marilyn Schlitz, vice-president for research and education at the Institute of Noetic Sciences and one of the scientists engaged in the earliest investigations of remote influence, defined intention as 'the projection of awareness, with purpose and efficacy, toward some object or outcome'.[9] To influence physical matter, they believed, thought had to be highly motivated and targeted.

In a series of remarkable experiments, these scientists provided evidence that thinking certain directed thoughts could affect one's own body, inanimate objects and virtually all manner of living things, from single-celled organisms to human beings.

Two of the major figures in this tiny subgroup were former dean of engineering Robert Jahn at the Princeton Anomalies Engineering Research (PEAR) laboratory a Princeton University and his colleague Brenda Dunne, who together created a sophisticated, scholarly research programme grounded in hard science.

Over 25 years, Jahn and Dunne led what became a massive international effort to quantify what is referred to as 'micro-psychokinesis', the effect of mind on random-event generators (REGs), which perform the electronic, twenty-first century equivalent of a toss of a coin.

The output of these machines (the computerized equivalent of heads or tails) was controlled by a randomly alternating frequency of positive and negative pulses. Because their activity was utterly random, they produced 'heads' and 'tails' each roughly 50 per cent of the time, according to the laws of probability.

The most common configuration of the REG experiments was a computer screen randomly alternating two attractive images – say, of cowboys and Indians. Participants in the studies would be placed in front of the computers and asked to try to influence the machine to produce more of one image – more cowboys, say – then to focus on producing more images of Indians, and then to try not to influence the machine in either direction.

Over the course of more than two and a half million trials Jahn and Dunne decisively demonstrated that human intention can influence these electronic devices in the specified direction,[10] and their results were replicated independently by 68 investigators.[11]

Experiments have conclusively shown that intention; directed thought, can absolute influence the physical world.

While PEAR concentrated on the effect of mind on inanimate objects and processes, many other scientists experimented with the effect of intention on living things.

A diverse number of researchers demonstrated that human intention can affect an enormous variety of living systems: bacteria, yeast, algae, lice, chicks, mice, gerbils, rats, cats and dogs.[12]

A number of these experiments have also been carried out with human targets; intention has been shown to affect many biological processes within the receiver, including gross motor movements and those in the heart, the eye, the brain and the respiratory system.

Animals themselves proved capable of acts of effective intention.

In one ingenious study by René Peoc'h of the Fondation ODIER in Nantes, France, a roboti 'mother hen', constructed from a moveable random-event generator, was 'imprinted' on a group of baby chicks soon after birth.

The robot was placed outside the chicks' cage, where it moved around freely, as its path was tracked and recorded.

Eventually, it was clear that the robot was moving towards the chicks two and a half times more often than it would ordinarily; the 'inferred intention' of the chicks – their desire to be close to their mother – appeared to affect the robot, drawing it closer to the cage.

In over 80 similar studies, in which a lighted candle was placed on a movable REG, baby chicks kept in the dark, finding the light comforting, managed to influence the robot to spend more time than normal in the vicinity of their cage.[13]

The largest and most persuasive body of research has been amassed by William Braud, a psychologist and the research director of the Mind Science Foundation in San Antonio, Texas, and, later, the Institute of Transpersonal Psychology. Braud and his colleagues demonstrated that human thoughts can affect the direction in which fish swim, the movement of other animals such as gerbils, and the breakdown of cells in the laboratory.[14]

Braud also designed some of the earliest well-controlled studies of mental influence on human beings. In one group of studies, Braud demonstrated that one person could affect the autonomic nervous system (or fight-or-flight mechanisms) of another. [15]

Which is one of the many reasons why I tell people that they must isolate themselves from chronically negative people, sociopaths, psychopaths, and people with social,

mental or emotional disorders. these individuals will absolutely affect your life, and often it is for their benefit, whatever they perceive it to be, and not yours.

Electrodermal activity (EDA) is a measure of skin resistance and shows an individual's state of stress; a change of EDA usually occurs if someone is stressed or made uncomfortable in some way.[16]

Braud's signature study tested the effect on EDA of being stared at, one of the simplest means of isolating the effect of remote influence on a human being. He repeatedly demonstrated that people were subconsciously aroused while they were being stared at.[17]

Perhaps the most frequently studied area of remote influence concerns remote healing.

Some 150 studies, of variable scientific rigor, have been carried out,[18] and one of the best designed was conducted by the late Dr Elisabeth Targ. During the height of the AIDS epidemic in the 1980s, she devised an ingenious, highly controlled pair of studies, in which some 40 remote healers across America were shown to improve the health of terminal AIDS patients, even though the healers had never met or been in contact with their patients.[19]

The use of thought and intention as a means to heal others over distances has been scientifically confirmed to be valid.

Even some of the most rudimentary mind-over-matter experiments have had tantalizing results.

One of the first such studies involved attempts to influence a throw of the dice. To date, some 73 studies have examined the efforts of 2500 people to influence more than two and a half million throws of the dice, with extraordinary success. When all the studies were analyzed together, and allowances made for quality or selective reporting, the odds of the results occurring by chance alone were 1076 (1 followed by 76 zeros) to one.[20]

There was also some provocative material about spoon bending, that perennial party trick made popular by psychic Uri Geller. John Hasted, a professor at Birkbeck College at the University of London, had tested this with an ingenious experiment involving children.

Hasted suspended latch keys from the ceiling and placed the children 3 to 10 feet away from their target key, so that they could have no physical contact. Attached to each key was a strain gauge, which would detect and register on a strip chart recorder any change in the key.

Hasted then asked the children to try to bend the suspended metal. During the sessions, he observed not only the keys swaying and sometimes fracturing, but also abrupt and enormous spikes of voltage pulses up to 10 volts – the very limits of the chart recorder. Even more compelling, when children had been asked to send their intention to several keys hung separately, the individual strain recorders noted simultaneous signals, as though the keys were being affected in concert.[21]

My first wife has a first cousin that could bend spoons, and she did it right there within inches of my nose. It was real, and pretty darn amazing.

Most intriguing, in much of the research on psychokinesis, mental influence of any variety had produced measurable effects, no matter how far the distance between the sender or what point in time he generated his intention. According to the experimental evidence, the power of thought transcended time and space.

By the time these revisionists were finished, they had torn up the rule book and scattered it to the four winds. Mind in some way appeared to be inextricably connected to matter and, indeed, was capable of altering it. Physical matter could be influenced, even irrevocably altered, not simply by force, but through the simple act of formulating a thought.

Nevertheless, the evidence from these frontier scientists left three fundamental questions unanswered.

Through what physical mechanisms do thoughts affect reality?

At the time of this writing, some highly publicized studies of mass prayer showed no effect.

- Were certain conditions and preparatory states of mind more conducive to success than others?
- How much power did a thought have, for good or ill?
- How much of our lives could a thought actually change?

Most of the initial discoveries about consciousness occurred more than 30 years ago. More recent discoveries in frontier quantum physics and in laboratories around the globe offer answers to some of those questions. They provide evidence that our world is highly malleable, open to constant subtle influence. Recent research demonstrates that living things are constant transmitters and receivers of measurable energy. New models of consciousness portray it as an entity capable of trespassing physical boundaries of every description.

Intention appears to be something akin to a tuning fork, causing the tuning forks of other things in the universe to resonate at the same frequency.

The latest studies of the effect of mind on matter suggest that intention has variable effects that depend on the state of the host, and the time and the place where it originates. Intention has already been employed in many quarters to cure illness, alter physical processes and influence events.

It is not a special gift but a learned skill, readily taught. Indeed, we already use intention in many aspects of our daily lives.

Intention is a learned skill, and with practice, anyone can become proficient with it. A body of research also suggests that the power of an intention multiplies, depending upon how many people are thinking the same thought at the same time.[22]

The Intention Experiment consists of three aspects.

The **main body of the book (chapters 1–12)** attempts to synthesize all the experimental evidence that exists on intention into a coherent scientific theory of how intention works, how it can be used in your life and which conditions optimize its effect.

The **second portion of the book (chapter 13)** offers a blueprint for using intention effectively in your own life through a series of exercises and recommendations for how best to ‘power up’. This portion is also an exercise in frontier science. I am not an expert in human potential, so this is not a self-help manual, but a journey of discovery for me as well as you. I have extrapolated this programme from scientific evidence describing those circumstances that created the most positive results in psychokinetic laboratory experiences. We know for certain that these techniques have generated success under controlled experimental laboratory conditions, but I cannot guarantee they will work in your life. By making use of them, you will, in effect, engage in an ongoing personal experiment.

The **final section of the book (chapters 14 and 15)** consists of a series of personal and group experiments. Chapter 14 outlines a series of informal experiments on the use of intention in your own life for you to carry out individually. These mini ‘experiments’ are also intended to be pieces of research. You will have the opportunity to post your results on our website and share them with other readers.

Besides these individual experiments, I have also designed a series of large group experiments to be carried out by the readers of this book (chapter 15). With the aid of our highly experienced scientific team, *The Intention Experiment* will conduct periodic large-scale experiments to determine whether the focused intention of its readers has an effect on scientifically quantifiable targets.

All it requires is that you read the book, digest its contents, log on to the website (www.theintentionexperiment.com) and, after following the instructions and exercises at the back of this book, send out some highly specific thoughts, as and when described on the site. The first such studies will be carried out by the German physicist Fritz-Albert Popp, vice-president of the International Institute of Biophysics in Neuss, Germany (www.lifescientists.de), and his team of seven, psychologist Gary Schwartz and his colleagues at the University of Arizona at Tucson, and Marilyn Schlitz and Dean Radin of the Institute of Noetic Sciences.

Website experts have collaborated with our scientific team to design log-on protocols to enable us to identify which characteristics of a group or aspects of their thoughts produce the most effective results. For each intention experiment, a target will be selected – a specific living thing or a population where change caused by group intention can be measured. We have started with algae, the lowliest of subjects (see chapter 12), and, with every experiment, we will move on to an increasingly complex living target.

Our plans are ambitious: to tackle a number of societal ills. One eventual human target might be patients with a wound. It is known and accepted that wounds generally heal at a particular, quantifiable rate with a precise pattern.[23] Any departure from the norm can be precisely measured and shown to be an experimental effect. In that instance, our aim would be to determine whether focused group intention will enable wounds to heal more quickly than usual.

Naturally, you don't have to participate in our experiments. If you don't wish to get involved, you can read about the intention experiments of others, and use some of that information to inform how you use intention in your life.

Please do not casually participate in the experiments. In order for the experiment to work properly, you must read the book and digest its contents fully beforehand. The experimental evidence suggests that those who are the most effective have trained their minds, much as athletes train their muscles, to maximize their chances of success.

In order to discourage uncommitted participation, *The Intention Experiment* website contains a complicated password comprising some words or ideas from the book (which will change slightly every few months). In order to be part of the experiment, you will have to log on with the password and you will have to have read the book and understood it.

The website (www.theintentionexperiment.com) has a running clock (set to US Eastern Standard Time and Greenwich Mean Time). At a particular moment on a date

specified on the website, you will be asked to send a carefully worded, detailed intention, depending on the target site.

Once finished, the results of the experiments will be analysed and data-crunched by our scientific team, examined by a neutral statistician, and then published on the website and in subsequent printings of this book. The website will thus become the living sequel to the book you are holding in your hands. You simply need to consult the website periodically for announcements of the date of every experiment.

Hundreds of well-designed studies of group intention and remote mental influence have demonstrated significant results. Nevertheless, it might be the case that our experiments will not produce demonstrable, measurable effects, at first or indeed ever. As reputable scientists and objective researchers, we are duty-bound to report the data we have. As with all science, failure is instructive, helping us to refine the design of the experiments and the premises that they are based upon.

As you read this book, keep in mind that this is a work of frontier science. Science is a relentless process of self-correction. Assumptions originally considered as fact must often ultimately be discarded. Many – indeed, most – of the conclusions drawn in this book are bound to be amended or refined at a later date.

By reading this book and participating in its experiments you may well contribute to the world's knowledge, and possibly further a paradigm shift in our understanding of how the world works. Indeed, the power of mass intention may ultimately be the force that shifts the tide towards repair and renewal of the planet. When combined with hundreds of thousands of others, your solitary voice, now one barely audible note, could transmute into a thunderous symphony.

My own motive for writing *The Intention Experiment* was to make a statement about the extraordinary nature and power of consciousness. It may prove true that a single collective, directed thought is all it takes to change the world.

Notes – Introduction

1. For a complete description of these scientists and their findings, consult L. McTaggart, *The Field: the Quest for the Secret Force of the Universe*, London: HarperCollins, 2001.
2. The full title of Newton's major treatise is *Philosophiae Naturalis Principia Mathematica*, a name that offers a nod to its philosophical implications, although it is always referred to reverentially as the *Principia*.
3. R. P. Feynman, *Six Easy Pieces: The Fundamentals of Physics Explained*, London: Penguin, 1995: 24.

4. McTaggart, *The Field*, op. cit.
5. Eugene Wigner, the Hungarian-born American physicist who received a Nobel Prize for his contribution to the theory of quantum physics, is one of the early pioneers of the central role of consciousness in determining reality and argued, through a thought experiment called 'Wigner's friend', that the observer, 'the friend', might collapse Schrödinger's famous cat into a single state or, like the cat itself, remain in a state of superposition until another 'friend' comes into the lab. Other proponents of 'the observer effect' include John Eccles and Evan Harris Walker. John Wheeler is credited with espousing the theory that the universe is participatory: it only exists because we happen to be looking at it.
6. McTaggart, *The Field*, op. cit.
7. E. J. Squires, 'Many views of one world – an interpretation of quantum theory', *European Journal of Physics*, 1987; 8: 173.
8. B. F. Malle et al., *Intentions and Intentionality: Foundations of Social Cognition*, Cambridge, Mass.: MIT Press, 2001.
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12. These studies are itemized in great detail in D. Benor, *Spiritual Healing*, Volume 1, Southfield, Mich.: Vision Publications, 1992.
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18. W. Braud et al., 'Further studies of autonomic detection of remote staring: replication, new control procedures and personality correlates', *Journal of Parapsychology*, 1993; 57: 391–409; M. Schlitz and S. LaBerge 'Autonomic detection of remote observation; two conceptual replications', in D. Bierman (ed.), *Proceedings of Presented Papers: 37 Annual Parapsychological Association Convention*, Amsterdam, Fairhaven, Mass.: Parapsychological Association, 1994: 465–78.
19. D. Benor, *Spiritual Healing: Scientific Validation of a Healing Revolution*, Southfield, Mich.: Vision Publications, 2001.
20. F. Sicher, E. Targ et al., 'A randomized double-blind study of the effect of distant healing in a population with advanced AIDS: report of a small scale study', *Western Journal of Medicine*, 1998; 168 (6): 356–63. For a full description of the studies, see McTaggart, *The Field*, op. cit.: 181–96.
21. Psychologist Dean Radin conducted a meta-analysis in 1989 at Princeton University of all known dice experiments (73) published between 1930 and 1989. They are recounted in his book *Entangled Minds*, New York: Paraview, 2006: 148–51.
22. J. Hasted, *The Metal Benders*, London: Routledge & Kegan Paul, 1981, as cited in W. Tiller, *Science and Human Transformation; Subtle Energies Intentionality and Consciousness*, Walnut Creek, Calif.: Pavior Publications, 1997: 13.
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The Science of Intention

A human being is part of the whole, called by us 'universe', a part limited in time and space. He experiences himself, his thoughts and feelings as something separated from the rest – a kind of optical delusion of his consciousness.

-Albert Einstein

Chapter 1

Mutable Matter

FEW PLACES IN THE GALAXY are as cold as the helium-diluti refrigerator in Tom Rosenbaum's lab. Temperatures in the refrigerator – a boiler- sized circular apparatus with a number of cylinders – can descend to a few thousandths of a degree above absolute zero, almost 273°C below freezing – three thousand times colder than the farthest reaches of outer space. For two days, liquid nitrogen and helium circulate around the refrigerator, and then three pumps constantly blasting out gaseous helium

take the temperature down to the final rung. Without heat of any description, the atoms in matter slow to a crawl. At this scale of coldness, the universe would grind to a halt. It is the scientific equivalent of hell freezing over.

Absolute zero is the preferred temperature of a physicist like Tom Rosenbaum. At 47, as a distinguished professor of physics at the University of Chicago and former head of the James Franck Institute, Rosenbaum was in the vanguard of experimental physicists who liked exploring the limits of disorder in condensed-matter physics, the study of the inner workings of liquids and solids when their underlying order was disturbed.[1]

In physics, if you want to find out how something behaves, the best way is simply to make it uncomfortable and then see what happens. Creating disorder usually involves adding heat or applying a magnetic field to determine how it will react when disturbed and also to determine which spin position – or magnetic orientation – the atoms will choose.

Most of his colleagues in condensed-matter physics remained interested in symmetrical systems such as crystalline solids, whose atoms are arranged in orderly array, like eggs in a carton, but Rosenbaum was drawn to strange systems that were inherently disordered – to which more conventional quantum physicists referred disparagingly as ‘dirt’.

In dirt, he believed, lay exposed the unprobed secrets of the quantum universe, uncharted territory that he was happy to navigate.

He loved the challenge posed by spin glasses, strange hybrids of crystals, with magnetic properties, technically considered slow-moving liquids. Unlike a crystal, whose atoms point in the same direction in perfect alignment, the tiny magnets associated with the atoms of a spin glass are wayward and frozen in disarray.

The use of extreme coldness allowed Rosenbaum to slow down the atoms of these strange compounds enough to observe them minutely, and to tease out their quantum mechanical essence. At temperatures near to absolute zero, when their atoms are nearly stationary, they begin taking on new collective properties.

Rosenbaum was fascinated by the recent discovery that systems disorderly at room temperature display a conformist streak once they are cooled down. For once, these delinquent atoms begin to act in concert.

Examining how molecules behave as a group in various circumstances is highly instructive about the essential nature of matter.

In my own journey of discovery, Rosenbaum's laboratory seemed the most appropriate place to begin. There, at those lowest temperatures where everything occurs in slow motion, the true nature of the most basic constituents of the universe might be revealed. I was looking for evidence of ways in which the components of our physical universe, which we think of as fully realized, are capable of being fundamentally altered.

I also wondered whether it could be shown that quantum behaviour like the observer effect occurs outside the subatomic world, in the world of the everyday. What Rosenbaum had discovered in his refrigerator might offer some vital clues as to how every object or organism in the physical world, which classical physics depicts as an irreversible fact, a finalized assemblage only changeable by the brute force of Newtonian physics, could be affected and ultimately altered by the energy of a thought.

According to the second law of thermodynamics, all physical processes in the universe can only flow from a state of greater to lesser energy. We throw a stone into a river and the ripple it makes eventually stops. A cup of hot coffee left standing can only grow cold.

Things inevitably fall apart; everything travels in a single direction, from order to disorder.

But this might not always be inevitable, Rosenbaum believed. Recent discoveries about disordered systems suggested that certain materials, under certain circumstances, might counteract the laws of entropy and come together rather than fall apart.

Was it possible that matter could go in the opposite direction, from disorder to greater order?

For ten years Rosenbaum and his students at the James Franck Institute had been asking that question of a small chunk of lithium holmium fluoride salt. Inside Rosenbaum's refrigerator lay a perfect chip of rose-coloured crystal, no bigger than the head of a pencil, wrapped in two sets of copper coils.

Over the years, after many experiments with spin glasses, Rosenbaum had grown very fond of these dazzling little specimens, one of the most naturally magnetic substances on earth. This characteristic presented the perfect situation in which to study disorder, but only after he had altered the crystal beyond recognition into a disordered substance.

He had first instructed the laboratory that grew the crystals to combine the holmium with fluorine and lithium, the first metal on the periodic table. The resulting lithium holmium fluoride salt was compliant and predictable – a highly ordered substance whose atoms behaved like a sea of microscopic compasses all pointing north.

Rosenbaum then had wreaked havoc on the original salt compound, instructing the lab to rip out a number of the atoms of holmium, bit by bit, and replace them with yttrium, a silvery metal without such natural magnetic attraction, until he was left with a strange hybrid of a compound: a salt called lithium holmium yttrium tetrafluoride.

By virtually eliminating the magnetic properties of the compound, Rosenbaum eventually had created spin-glass anarchy – the atoms of this Frankenstein monstrosity pointing any way they liked. Being able to manipulate the essential property of elements like holmium by creating weird new compounds so cavalierly was a little like having ultimate control over matter itself. With these new spin-glass compounds, Rosenbaum could virtually change the properties of the compound at will; he could make the atoms orientate in a particular direction, or freeze them in some random pattern.

Nevertheless, his omnipotence had a limit. Rosenbaum's holmium compounds behaved themselves in some regards, but not in others. One thing he could not do was to get them to obey the laws of temperature. No matter how cold Rosenbaum made his refrigerator, the atoms inside them resisted any sort of ordered orientation, like an army refusing to march in step.

If Rosenbaum was playing God with his spin glasses the crystal was Adam, stubbornly refusing to obey His most fundamental law.

Sharing Rosenbaum's curiosity about the strange property of the crystal compound was a young student called Sayantani Ghosh, one of his star PhD candidates. Sai, as her friends called her, a native of India, had graduated with a first-class honours degree from Cambridge, after which she had chosen Tom's lab for her doctoral programme in 1999. Almost immediately, she had distinguished herself by winning the Gregor Wentzel Prize, given each year by the University of Chicago's physics department to the best first-year graduate student teaching assistant. The slight 23-year-old, who at first glance appeared abashed, hiding behind her copious dark hair, had soon impressed her peers and teachers alike with her bold authority, a rarity among science students, and her ability to translate complex ideas to the level an undergraduate could comprehend. Sai shared the distinction of winning the coveted prize with only one other woman since its inception 25 years before.

According to the laws of classical physics, applying a magnetic field will disrupt the magnetic alignment of a substance's atoms. The degree to which this happens is the

salt's 'magnetic susceptibility'.

The usual pattern with a disordered substance is that it will respond to the magnetic field for a time and then plateau and tail off, as the temperature drops or the magnetic field reaches a point of magnetic saturation. The atoms will no longer be able to flip in the same direction as that of the magnetic field and so will begin to slow down.

In Sai's first experiments, the atoms in the lithium holmium yttrium salt, as predicted, grew wildly excited with the application of the magnetic field. But then, as Sai increased the field, something strange began to happen. The more she turned up the frequency, the faster the atoms continued to flip over.

What is more, all the atoms, which had been in a state of disarray, began pointing in the same direction and operating as a collective whole. Then, small clusters of about 260 atoms aligned, forming 'oscillators', spinning collectively in one direction or another.

No matter how strong the magnetic field that Sai applied, the atoms remained stubbornly aligned with each other, acting in concert. This self-organization persisted for 10 seconds.

At first, Sai and Rosenbaum thought these effects might have something to do with the strange effects of the remaining atoms of holmium, known to be one of the very few substances in the world with such long-range internal forces that in some quarters it was described and worked out mathematically as something existing in another dimension.[2] Although they didn't understand the phenomenon they had observed, they wrote up their results, which were published in the journal *Science* in 2002.[3]

Rosenbaum decided to carry out another experiment to attempt to isolate the property in the crystal's essential nature that had enabled it to override such strong outside influences. He left the study's design to his bright young graduate student, suggesting only that she create a computerized three-dimensional mathematical simulation of the experiment she had intended to carry out. In experiments of this nature on such tiny matter, physicists must rely on a computerized simulation to confirm mathematically the reactions they are witnessing experimentally.

Sai spent months developing the computer code and building her simulation. The plan was to find out a bit more about the salt's magnetic capability, by applying two systems of disorder to the crystal chip: higher temperatures and a stronger magnetic field.

She prepared the sample by placing it in a little 2.4 x 4.8 cm copper holder, then wrapped two coils around the tiny crystal: one a gradiometer, to measure its magnetic susceptibility and the direction of spin of the individual atoms, and the other to cancel out any random flux affecting the atoms inside.

A connection attached to her PC would enable her to change the voltage, the magnetic field or the temperature, and would record any changes whenever she altered one of the variables by the tiniest degree.

She began lowering the temperature, a fraction of a kelvin (K) at a time, and then began applying a stronger magnetic field. To her amazement, the atoms kept aligning progressively. Then she tried applying heat, and discovered they again aligned. No matter what she did, in every instance the atoms ignored the outside interference. Although she and Tom had flushed out most of the compound's magnetic component, of its own volition, as it were, it was turning into a larger and larger magnet.

That's weird, she thought. Perhaps she should take more data, just to ensure they had encountered nothing strange in the system.

She repeated her experiment over six months until the early spring of 2002, when her computer simulation was finally complete. One evening, she mapped the results of the simulation on a graph, and then she superimposed the results from her actual experiment.

It was as though she had drawn a single line.

There on the computer screen was a perfect duplicate: the diagonal line formed from the computer simulation lay exactly over the diagonal line created from the results of the experiment itself.

What she had witnessed in the little crystal was not an artefact, but something real that she had now reproduced in her computer simulation. She had even mapped out where the atoms should have been on the graph, had they been obeying the usual laws of physics.

But there they were in a line: a law completely unto themselves.

She wrote Rosenbaum a guarded email late that evening:

'I've got something interesting to show you in the morning.'

The following day, they examined her graph. There was no other possibility, they both realized; the atoms had been ignoring her and instead were controlled by the activity of their neighbors. No matter whether she blasted the crystal with a strong magnetic field or an increase in temperature, the atoms overrode this outside disturbance.

The only explanation was that the atoms in the sample crystal were internally organizing and behaving like one single giant atom. All the atoms, they realized with some alarm, must be entangled.

One of the strangest aspects of quantum physics is a feature called 'non- locality', also poetically referred to as 'quantum entanglement'. The Danish physicist Niels Bohr discovered that once subatomic particles such as electrons or photons are in contact, they remain cognizant of and influenced by each other instantaneously over any distance forever, despite the absence of the usual things that physicists understand are responsible for influence, such as an exchange of force or energy.

When entangled, the actions – for instance, the magnetic orientation – of one will always influence the other in the same or the opposite direction, no matter how far they are separated. Erwin Schrödinger, another one of the original architects of quantum theory, believed that the discovery of non-locality represented no less than quantum theory's defining moment – its central property and premise.

The activity of entangled particles is analogous to a set of twins being separated at birth, but retaining identical interests and a telepathic connection forever. One lives in Colorado, and the other in London. Although they never meet again, both like the color blue. Both take a job in engineering. Both like to ski; in fact when one falls down and breaks his right leg at Vale, his twin breaks his right leg at precisely that moment, even though he is 4000 miles away, sipping a latte at Starbucks.[4]

Albert Einstein refused to accept non-locality, referring to it disparagingly as '*spukhafte Fernwirkungen*' or 'spooky action at a distance'.

This type of instantaneous connection would require information traveling faster than the speed of light, he argued through a famous thought experiment, which would violate his own special relativity theory.[5]

Since the formulation of Einstein's theory, the speed of light (299,792,458 meters per second) has been used as the absolute limiting factor on how quickly one thing can affect something else. Things are not supposed to be able to affect other things faster than the time it would take the first thing to travel to the second thing at the speed of light.

Nevertheless, modern physicists, such as Alain Aspect and his colleagues in Paris, have demonstrated decisively that the speed of light is not an absolute outer boundary in the subatomic world.

Aspect's experiment, which concerned two photons fired off from a single atom, showed that the measurement of one photon instantaneously affected the position of

the second photon[6] so that it has the same or opposite spin or position (as IBM physicist Charles H. Bennett once put it, 'opposit luck').[7]

The two photons continued to talk to each other and whatever happened to one was identical to, or the very opposite of, what happened to the other. Today, even the most conservative physicists accept non-locality as a strange feature of subatomic reality. [8]

Most quantum experiments incorporate some test of Bell's Inequality. This famous experiment in quantum physics was carried out by John Bell, an Irish physicist who developed a practical means to test how quantum particles really behaved.[9]

This simple test required that you get two quantum particles that had once been in contact, separate them and then take measurements of the two. It is analogous to a couple named Daphne and Ted who have once been together but are now separated. Daphne can choose one of two possible directions to go in and so can Ted. According to our commonsense view of reality, Daphne's choice should be utterly independent of Ted's.

When Bell carried out his experiment, the expectation was that one of the measurements would be larger than the other – a demonstration of 'inequality'. However, a comparison of the measurements showed that both were the same and so his inequality was 'violated'.

Some invisible wire appeared to be connecting these quantum particles across space, to make them follow each other. Ever since, physicists have understood that when a violation of Bell's Inequality occurs, it means that two things are *entangled*.

Bell's Inequality has enormous implications for our understanding of the universe.

By accepting non-locality as a natural facet of nature we are acknowledging that two of the bedrocks on which our world view rests are wrong: that influence only occurs over time and distance, and that particles like Daphne and Ted, and indeed the things that are made up of particles, only exist independently of each other.

Although modern physicists now accept non-locality as a given feature of the quantum world, they console themselves by maintaining that this strange, counter-intuitive property of the subatomic universe does not apply to anything bigger than a photon or electron.

Once things got to the level of atoms and molecules, which in the world of physics is considered 'macroscopic', or large, the universe started behaving itself again, according to predictable, measurable, Newtonian laws.

With one tiny thumbnail's worth of crystal, Rosenbaum and his graduate student demolished that delineation.

They had demonstrated that big things like atoms were non-locally connected, even in matter so large you could hold it in your hand. Never before had quantum non-locality been demonstrated on such a scale. Although the specimen had been only a tiny chip of salt, to the subatomic particle, it was a palatial country mansion, housing a billion billion (1,000,000,000,000,000,000 or 10^{18}) atoms.

Rosenbaum, ordinarily loathe to speculate about what he could not yet explain, realized that they had uncovered something extraordinary about the nature of the universe.

And I realized they had discovered a mechanism for intention: they had demonstrated that atoms, the essential constituents of matter, could be affected by non-local influence. Large things like crystals were not playing by the grand rules of the game, but by the anarchic rules of the quantum world, maintaining invisible connections without obvious cause.

In 2002, after Sai wrote up their findings, Rosenbaum polished up the wording and sent off their paper to *Nature*, a journal notorious for conservatism and exacting peer review. After four months of responding to the suggestions of reviewers, Ghosh finally got her paper published in the world's premier scientific journal, a laudatory feat for a 26-year-old graduate student.[10]

One of the reviewers, Vlatko Vedral, noted the experiment with a mix of interest and frustration.[11] A Yugoslav who had studied at Imperial College, London, during his country's civil war and subsequent collapse, Vedral had distinguished himself in his adopted country and been chosen to head up quantum information science at the University of Leeds. Vedral, who was tall and leonine, was part of a small group in Vienna working on frontier quantum physics, including entanglement.

Vedral first theoretically predicted the effect that Ghosh and Rosenbaum eventually found three years later. He had submitted the paper to *Nature* in 2001, but the journal, which preferred experiment to theory, had rejected it. Eventually, Vedral managed to publish his paper in *Physical Review Letters*, the premier physics journal.[12] After *Nature* decided to publish Ghosh's study, its editors threw him a conciliatory bone. They allowed him to be a reviewer on the paper, and then offered him a place in the same issue to write an opinion piece on the findings.

In the article, Vedral allowed himself some speculation. Quantum physics is accepted as the most accurate means of describing how atoms combine to form molecules, he

wrote, and since molecular relationship is the basis of all chemistry, and chemistry is the basis of biology, the magic of entanglement could well be the key to life itself.[13]

Vedral and a number of others in his circle did not believe that this effect was unique to holmium. The central problem in uncovering entanglement is the primitive state of our technology; isolating and observing this effect is only possible at the moment by slowing atoms down so much in such cold conditions that they are hardly moving. Nevertheless, a number of physicists had observed entanglement in matter at 200 K, or -73°C – a temperature that can be found on Earth in some of its very coldest places.

Other researchers have proved mathematically that everywhere, even inside of our own bodies, atoms and molecules are engaged in an instantaneous and ceaseless passing back and forth of information.

Thomas Durt of Vrije University in Brussels demonstrated through elegant mathematical formulations that almost all quantum interactions produce entanglement, no matter what the internal or surrounding conditions. Even photons, the tiniest particles of light emanating from stars, are entangled with every atom they meet on their way to earth.[14]

Entanglement at normal temperatures appears to be a natural condition of the universe, even in our bodies. Every interaction between every electron inside of us creates entanglement. According to Benni Reznik, a theoretical physicist at Tel Aviv University in Israel, even the empty space around us is heaving with entangled particles.[15]

The English mathematician Paul Dirac, an architect of quantum field theory, first postulated that there is no such thing as nothingness, or empty space. Even if you tipped all matter and energy out of the universe and examined all the ‘empty’ space between the stars you would discover a netherworld world teeming with subatomic activity.

In the world of classical physics, a field is a region of influence, in which two or more points are connected by a force, like gravity or electromagnetism. However, in the world of the quantum particle, fields are created by exchanges of energy.

According to Heisenberg’s uncertainty principle, one reason that quantum particles are ultimately unknowable is because their energy is always being redistributed in a dynamic pattern. Although often rendered as tiny billiard balls, subatomic particles more closely resemble little packets of vibrating waves, passing energy back and forth as if in an endless game of basketball. All elementary particles interact with each other by exchanging energy through what are considered temporary or ‘virtual’

quantum particles. These are believed to appear out of nowhere, combining and annihilating each other in less than an instant, causing random fluctuations of energy without any apparent cause. Virtual particles, or negative energy states, do not take physical form, so we cannot actually observe them. Even 'real' particles are nothing more than a little knot of energy, which briefly emerge and disappear back into the underlying energy field.

These back-and-forth passes, which rise to an extraordinarily large ground state of energy, are known collectively as the Zero Point Field.

Zero point field is the basis of many types of substantive extraterrestrial technology. The field is called 'zero point' because even at temperatures of absolute zero, when all matter theoretically should stop moving, these tiny fluctuations are still detectable. Even at the coldest place in the universe, subatomic matter never comes to rest, but carries on this little energy tango.[16]

The energy generated by every one of these exchanges between particles is unimaginably tiny – about half a photon's worth. However, if all exchanges between all subatomic particles in the universe were to be added up, it would produce an inexhaustible supply of energy of unfathomable proportions, exceeding all energy in matter by a factor of 10^{40} , or 1 followed by 40 zeros.[17] Richard Feynman himself once remarked that the energy in a cubic meter of space was enough to boil all the oceans of the world.[18]

After the discoveries of Heisenberg about Zero Point energy, most conventional physicists have subtracted the figures symbolizing Zero Point energy from their equations. They assumed that, because the Zero Point Field was ever present in matter, it did not change anything and so could be safely 'renormalized' away.

However, in 1973, when trying to work out an alternative to fossil fuel during the petrol crisis, American physicist Hal Puthoff, inspired by the Russian Andrei Sakharov, began trying to figure out how to harness the teeming energy of empty space for transport on earth and to distant galaxies.

Puthoff spent more than 30 years examining the Zero Point Field.

With some colleagues, he had proved that this constant energy exchange of all subatomic matter with the Zero Point Field accounts for the stability of the hydrogen atom, and, by implication, the stability of all matter.[19]

Remove the Zero Point Field and all matter would collapse in on itself.

He also demonstrated that Zero Point energy is responsible for two basic properties of mass: inertia and gravity.[20]

Puthoff also worked on a multimillion-dollar project funded by Lockheed Martin and a variety of American universities, to develop Zero Point energy for space travel – a programme that finally went public in 2006.

Many strange properties of the quantum world, like uncertainty or entanglement, could be explained if you factored in the constant interaction of all quantum particles with the Zero Point Field. To Puthoff, science's understanding of the nature of entanglement was analogous to two sticks stuck in the sand at the edge of the ocean, about to be hit by a huge wave. If they both were knocked over, and you did not know about the wave, you would think that one stick was affecting the other and call it a non-local effect. The constant interaction of quantum particles with the Zero Point Field might be the underlying mechanism for non-local effects between particles, allowing one particle to be in touch with every other particle at any moment.[21]

Benni Reznik's work in Israel with the Zero Point Field and entanglement began mathematically with a central question: what would happen to a hypothetical pair of probes interacting with the Zero Point Field? According to his calculations, once they began interacting with the Zero Point Field, the probes would begin talking to each other and ultimately become entangled.[22]

If all matter in the universe were interacting with the Zero Point Field, it meant quite simply, that all matter was interconnected and potentially entangled throughout the cosmos through quantum waves.[23]

And if we and all of empty space are a mass of entanglement, we must be establishing invisible connections with things at a distance to ourselves.

Acknowledging the existence of the Zero Point Field and entanglement offers a ready mechanism for why signals being generated by the power of thought can be picked up by someone else many miles away.

* * *

Sai Ghosh had proved that non-locality existed in the large building blocks of matter and the other scientists proved that all matter in the universe was, in a sense, a satellite of a large central energy field. But how could matter be affected by this connection? The central assumption of all of classical physics is that large material things in the universe are set pieces, a *fait accompli* of manufacture.

How can they possibly be changed?

Vedral had an opportunity to examine this question when he was invited to work with the renowned quantum physicist Anton Zeilinger. Zeilinger's Institute for Experimental Physics lab at the University of Vienna was at the very frontier of some of the most

exotic research into the nature of quantum properties. Zeilinger himself was profoundly dissatisfied with the current scientific explanation of nature, and he had passed on that dissatisfaction and the quest to resolve it to his students.

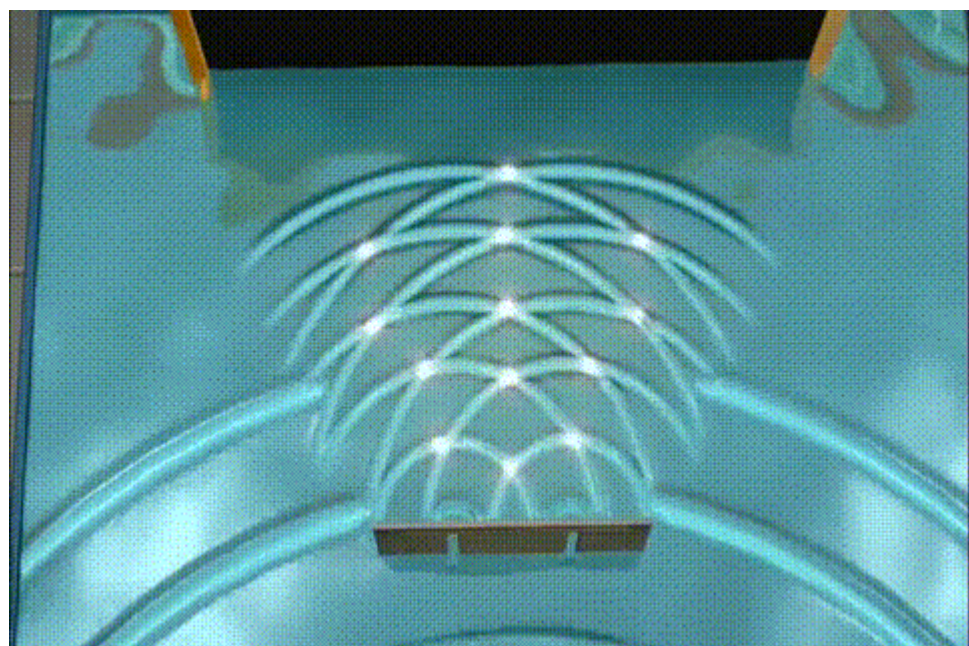
In a flamboyant gesture, Zeilinger and his team had entangled a pair of photons from beneath the River Danube. They had set up a quantum channel via a glass fibre and run it across the river bed of the Danube. In his lab, Zeilinger liked to refer to individual photons as Alice and Bob, and sometimes, if he needed a third photon, Carol or Charlie. Alice and Bob, separated by 600 metres of river and nowhere in sight of each other, maintained a non-local connection.[24]

Zeilinger was particularly interested in superposition, and the implications of the Copenhagen Interpretation – that subatomic particles exist only in a state of potential.

Could objects, and not simply the subatomic particles that compose them, he wondered, exist in this hall-of-mirrors state?

To test this question, Zeilinger employed a piece of equipment called a Talbot Lau interferometer, developed by some colleagues at MIT, using a variation on the famous double-slit experiment of Thomas Young, a British physicist of the nineteenth century. In Young's experiment, a beam of pure light is sent through a single hole, or slit, in a piece of cardboard, then passes through a second screen with two holes before finally arriving at a third, blank screen.

When two waves are in phase (that is, peaking and troughing at the same time), and bump into each other – technically called 'interference' – the combined intensity of the waves is greater than each



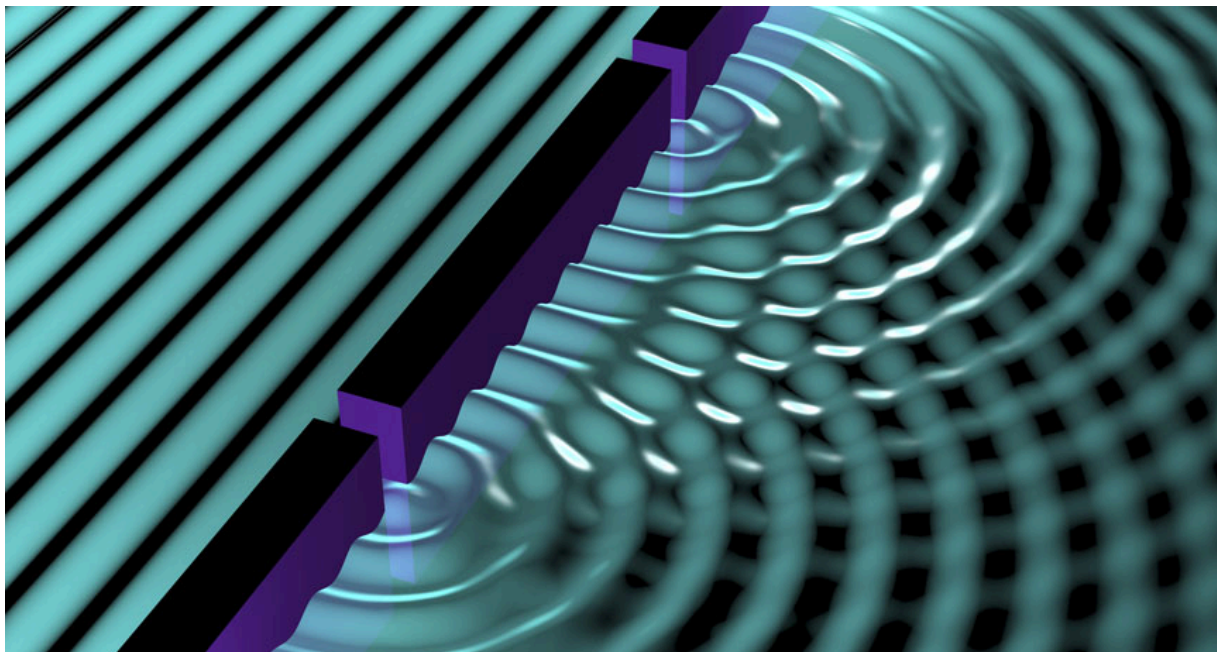
Young's experiment.

individual amplitude. The signal gets stronger. This amounts to an imprinting or

exchange of information, called 'constructive interference'. If one is peaking when the other troughs, they tend to cancel each other out – called 'destructive interference'. With constructive interference, when all the waves are wiggling in synch, the light will get brighter; destructive interference will cancel out the light and result in complete darkness.

In the experiment, the light passing through the two holes forms a zebra pattern of alternating dark and light bands on the final blank screen. If light were simply a series of particles, two of the brightest patches would appear directly behind the two holes of the second screen. However, the brightest portion of the pattern is halfway between the two holes, caused by the combined amplitude of those waves that most interfere with each other. From this pattern, Young was the first to realize that light beaming through the two holes spreads out in overlapping waves.

A modern variation of the experiment fires off single photons through the double slit. These single photons also produce zebra patterns on the screen, demonstrating that even single units of light travel as a smeared-out wave with a large sphere of influence.



Young's experiment.

Twentieth-century physicists went on to use Young's experiment with other individual quantum particles, and held it up as proof that quantum physics had Through-the-Looking-Glass properties: *quantum entities acted wavelike and travelled through both slits at once*. Fire a stream of electrons at the triple screens, and you end up with the interference patterns of alternating light and dark patches, just as you do with a beam of light. Since you need at least two waves to create such interference patterns, the

implication of the experiment is that the photon is somehow mysteriously able to travel through both slits at the same time and interfere with itself when it reunites.

The double-slit experiment encapsulates the central mystery of quantum physics

the idea that a subatomic particle is not a single seat but the entire stadium. It also demonstrates the principle that electrons, which exist in a hermetic quantum state, are ultimately unknowable. You could not identify something about a quantum entity without stopping the particle in its tracks, at which point it would collapse to a single point.

In Zeilinger's adaptation of the slit experiment, using molecules instead of subatomic particles, the interferometer contained an array of slits in the first screen, and a grating of identical parallel slits in the second one, whose purpose was to diffract (or deflect) the molecules passing by. The third grating, turned perpendicular to the beam of molecules, acted as a scanning 'mask', with the ability to calculate the size of the waves of any of the molecules passing through, by means of a highly sensitive laser detector to locate the positions of the molecules and their interference patterns.

For the initial experiment, Zeilinger and his team carefully chose a batch of fullerene molecules, or 'buckyballs' made of 60 carbon atoms. At one nanometre apiece, these are the behemoths of the molecular world. They selected fullerene not only for its size but also for its neat arrangement, with a shape like a tiny symmetrical football.

It was a delicate operation. Zeilinger's group had to work with just the right temperature; heating the molecules just a hair too much would cause them to disintegrate. Zeilinger heated the fullerenes to 900 K so they would create an intense molecular beam, then fired them through the first screen; they then passed through the second screen before making a pattern on the final screen. The results were unequivocal. Each molecule displayed the ability to create interference patterns with itself. Some of the largest units of physical matter had not 'localized' into their final state. Like a subatomic particle, these giant molecules had not yet gelled into anything real.

The Vienna team scouted out some other molecules that were double the size and oddly shaped to see if geometrically asymmetric molecules also demonstrated the same magical properties. They settled on gigantic fluorinated American football-shaped molecules of 70 carbon atoms and pancake-shaped tetraphenylporphyrin, a derivative of the biodye present in chlorophyll. At more than 100 atoms apiece, both of these entities are among the largest molecules on the planet. Again, each one created an interference pattern with itself.

Zeilinger's group repeatedly demonstrated that the molecules could be two places at once, which meant that they remained in a state of superposition even at this large scale.[25]

They had proved the unthinkable: the largest components of physical matter and living things exist in a malleable state.[26]

Sai Ghosh didn't often think about the implications of her discovery.

She was content with the knowledge that her experiment had made a very nice paper, and might help along her career as an assistant professor involved in research into miniaturization, the direction she believed quantum mechanics was heading. Occasionally, she allowed herself to speculate that her crystal might have proved something important about the nature of the universe. But she was only a postgraduate student. What did she, after all, really know about how the world worked?

But to me, Ghosh's research and Zeilinger's work on the double-slit experiment represent two defining moments in modern physics. Ghosh's experiments show that an invisible connection exists between the fundamental elements of matter, which is often so strong that it can override classical methods of influence, such as heat or a push. Zeilinger's work demonstrated something even more astonishing. Large matter was neither something solid and stable nor something that necessarily behaved according to Newtonian rules. Molecules needed some other influence to settle them into a completed state of being.

Theirs were the first evidence that the peculiar properties of quantum physics do not simply occur at the quantum level with subatomic particles, but also in the world of visible matter. Molecules also exist in a state of pure potential, not a final actuality. Under certain circumstances, they escape Newtonian rules of force and display quantum non-local effects. The fact that something as large as a molecule can become entangled suggests that there are not two rule books – the physics of the large and the physics of the small – but only a single rule book for all of life.

These two experiments also hold the key to a science of intention – how thoughts are able to affect finished, solid matter.

Thoughts create reality. They can change the physical world around us in the most profound manners. Thus we absolutely need to have direct and substantive control over our thoughts.

They suggest that the observer effect occurs not simply in the world of the quantum particle but also in the world of the everyday. Things no longer should be seen to

exist in and of themselves but, like a quantum particle, only in relationship. Co-creation and influence may be a basic, inherent property of life.

Our observation of every component in our world may help to determine its final state, which suggests that we are likely to be influencing every large thing we see around us.

When we enter a crowded room, when we engage with our partners and our children, when we gaze up at the sky, we may be creating and even influencing at every moment. We can't yet demonstrate this at normal temperatures; our equipment is still too crude. But we already have some preliminary proof: the physical world – matter itself – appears to be malleable, susceptible to influence from the outside.

Notes – Chapter 1: Mutable Matter

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The Human Antenna

IN 1951, AT THE AGE OF SEVEN, Gary Schwartz made a remarkable discovery. He had been trying to get a good picture on the family's television set. The recently acquired black and white Magnavox set encased behind the doors of its boxed walnut console fascinated him, not because of the people in the moving pictures so much as the means by which they arrived in his living room in the first place.

The mechanisms of the relatively new invention remained a mystery, even to most adults. Television, like any other electrical gadget, was something the precocious child longed to take apart and understand. This passion had already found expression with the worn-out radios given to him by his grandfather.

Ignatz Schwartz sold replacement tubes for televisions and radios in his drug store in Great Neck, Long Island, and those that were beyond repair were handed over to his grandson to disassemble. In a corner of Gary's bedroom lay a mass of experimental debris – tubes, resistors and the carcasses of radios heaped on the cosmetic display racks he had borrowed from his grandfather – the first signs of what would become a lifelong fascination with electronics.

Gary knew that the way you twisted the rabbit-ear antenna on top of the television would determine the clarity of the picture. His father had explained that television sets were powered by something invisible, similar to radio waves, that flew through the air and were somehow translated into an image.

Gary had even carried out some rudimentary experiments. When you stood somewhere between the antenna and the television, you could make the picture go away. When you touched the antenna in certain ways, you made the picture clearer.

One day, on a whim, Gary unscrewed the antenna and placed his finger on the screw where the cable had been. What had been a mass of squiggles and static noise on the screen suddenly coalesced into a perfect image.

Even at that young age, he had understood that he had witnessed something extraordinary about human beings: his body was acting like a television antenna, a receiver of this invisible information.

He tried the same experiment with a radio – substituting his finger for the antenna, and the same thing happened.

Something in the makeup of a person was not unlike the rabbit ears that helped produce his television image. He too was a *receiver* of invisible information, with the ability to pick up signals transmitted across time and space.

Until he was 15, however, he could not visualize what these signals were made of. He had learned to play the electric guitar and had often wondered what unseen influences allowed the instrument to create different sounds. He could play the same note, middle C, and yet produce more of a treble or bass sound, depending on which way he turned the knob. How was it possible that a single note could sound so different? For a science project, he created multiple-track recordings of his music and then located a company in upstate New York that had equipment designed to analyse the frequency of sound. When he fed his recordings into the equipment, it quickly deconstructed the notes down to their essence.

Each note registered as a batch of squiggles across the screen of the cathode-ray tube in front of him – a complex mix of hundreds of frequencies representing a blend

of overtones that would subtly change when he turned the knob to treble or bass. He knew that these frequencies were waves, represented on the monitor as a sideways S, or sine curve, like a skipping rope held at both ends and wriggled, and that they had periodic oscillations, or fluctuations, similar to the waves on Long Island Sound.

Every time he spoke, he knew he generated similar frequencies through his voice. He remembered his early television experiments and wondered whether a field of energy pulsed inside him and shared a kinship with sound waves.[1](#)

Gary's childhood experiments may have been rudimentary, but he had already stumbled across the central mechanism of intention. Something in the quality of our thoughts was a constant transmission, not unlike a television station.

As an adult, Schwartz, still a bustling dynamo of enthusiasms, found an outlet in psychophysiology, then a fledgling study of the effect of the mind on the body. By the time he had accepted a post at the University of Arizona, which was known for encouraging freedom of research among its faculty, he had grown fascinated by biofeedback and the ways in which the mind could control blood pressure and a variety of illnesses – and the powerful physical effect of different types of thoughts.[2](#)

One weekend in 1994, at a conference on the relationship between love and energy, he sat in on a lecture by physicist Elmer Green, one of the pioneers of biofeedback. Green, like Schwartz, had grown interested in the energy being transmitted by the mind. To examine this more closely, he had decided to study remote healers and to determine whether they sent out more electrical energy than usual while in the process of healing.

Green reported in his lecture that he had built a room whose four walls and ceiling were entirely made of copper, and were attached to microvolt electroencephalogram (EEG) amplifiers – the kind used to measure the electrical activity in the brain. Ordinarily, an EEG amplifier is attached to a cap with imbedded electrodes, each of which records separate electrical discharges from different places in the brain. The cap is placed on a person's head, and the electrical activity picked up is displayed on the amplifier. EEG amplifiers are extraordinarily sensitive, capable of picking up the most minute of effects – even one-millionth of a volt of electricity.

In remote healing, Green suspected that the signal produced was electrical and emanated from the healer's hands. The copper wall acted like a giant antenna, magnifying the ability to detect the electricity from the healers and enabling Green to capture it from five directions.

He discovered that, whenever a healer sent healing, the EEG amplifier often recorded it as a huge surge of electrostatic charge, the same kind of the build up and

discharge of electrons that occurs after you shuffle your feet along a new carpet and then touch a metal doorknob.[3](#)

In the early days of the copper wall experiment, Green had been faced with an enormous problem. Whenever a healer so much as wriggled a finger, patterns got recorded on an EEG amplifier. Green had had to work out a means of separating out the true effects of healing from this electrostatic noise. The only way to do so, as he saw it, was to have his healers remain perfectly still while they were sending out healing energy.

Schwartz listened to the talk with growing fascination. Green was discarding what might be the most interesting part of the data, he thought. *One man's noise was another man's signal.*

Does movement, even the physiology of your breathing, create an electromagnetic signal big enough to be picked up on a copper wall? Could it be that human beings were not only receivers of signals but also *transmitters*?

It made perfect sense that we transmitted energy. A great deal of evidence had already proved that all living tissue has an electric charge. Placing this charge in three-dimensional space caused an electromagnetic field that traveled at the speed of light. The mechanisms for the transmission of energy were clear, but what was unclear was the degree to which we sent out electromagnetic fields just by simple movements and whether our energy was being picked up by other living things.

Schwartz was itching to test this out for himself. After the conference, he contacted Green for advice about how to build his own copper wall. He rushed to Home Depot, which did not stock copper shielding but did have aluminum shielding, which could also act as a rudimentary antenna.

He purchased some two by fours, placed them on glass bricks so that they would be isolated from the ground, and used them to assemble a 'wall'. After he had attached the wall to an EEG amplifier, he began playing around with the effects of his hand, waving it back and forth above the box. As he suspected, the amplifier tracked the movement. His hand movements were generating signals.[4](#)

Schwartz began demonstrating these effects in front of his students in his faculty office, making use of a bust of Einstein for dramatic effect. With these experiments, he made use of an EEG cap, with its dozens of electrodes. When not picking up brain signals, the cap will register only noise on the amplifier.

During his experiments, Schwartz placed the EEG cap on his Einstein bust, and turned on just a single electrode channel on the top of the cap. Then he moved his hand

over Einstein's head. As though the great man had suddenly experienced a moment of enlightenment, the amplifier suddenly came alive and produced evidence of an electromagnetic wave.

But the signal, Schwartz explained to his students, was not a sudden brain wave emitted from the lifeless statue – only the tracking of the electromagnetic field produced by his arm's movement. It seemed indisputable: his body must be sending out a signal with every single flutter of his hand.

Schwartz got more creative with his experiments. When he tried the same gesture from three feet away, the signal diminished. When he placed the bust in a Faraday cage, an enclosure of tightly knit copper mesh that screens out electromagnetic fields, all effect disappeared. This strange energy resulting from movement had all the hallmarks of electricity: it decreased with distance, and was blocked by an electromagnetic shield.

At one point, Schwartz asked one of the students to stand with his left hand over Einstein's head, with his right arm extended towards Schwartz, who was sitting in a chair three feet away. Schwartz moved his arm up and down. To the amazement of the other students, Schwartz's movement was picked up by the amplifier. The signal had passed through Schwartz's body and travelled through the student. Schwartz was still generating the signal, but this time, the student had become the antenna, receiving the signal and transmitting it to the amplifier, which acted as another antenna.

Schwartz realized he had hit upon the most important point of all his research.

Simple movement generated electrical charge, but, more important, it created a relationship. Every movement we make appears to be felt by the people around us.

The implications were staggering.

What if he were admonishing a student? What might be the physical effect on the student of wagging his finger while shouting 'Don't do that'? The student might feel as if he were getting shot with a wave of energy. Some people might even have more powerful positive or negative charges than others. In Elmer Green's copper wall experiment, all sorts of equipment malfunctioned in the presence of Roslyn Bruyere, a famous healer.

Schwartz was onto something fundamental about the actual energy that human beings emit. Could the energy of thought have the same effect as the energy of movement outside the thinker's own body? Did thoughts also create a relationship

with the people around us? Every intention towards someone else might have its own physical counterpart, which would be registered by its recipient as a physical effect.

Like Schwartz, I suspected the energy generated by thoughts did not behave in the same way as the energy generated by movement. After all, the signal from movement decreased over distance, much like ordinary electricity. With healing, distance appeared to be irrelevant. The energy of intention, if indeed there were any, would have to be more fundamental than that of ordinary electromagnetism – and lie somewhere, perhaps, in the realm of quantum physics. How could I test the energetic effects of intention? Healers, who appeared to be sending more energy than normal through their healing, offered an obvious place to start.

Elmer Green demonstrated in his research that an enormous surge of electrostatic energy occurred during healing. When a person is simply standing still, his or her breathing and beating heart will produce electrostatic energy of 10–15 millivolts on the EEG amplifiers; during activities requiring focused attention, such as meditation, the energy will surge up to 3 volts. During healing, however, Green's healers produced voltage surges up to 190 volts; one produced 15 such pulses, which were 100,000 times higher than normal, with smaller pulses of 1–5 volts appearing on each of the four copper walls. On investigating the source of this energy, Green discovered that the pulses were coming from the healer's abdomen, called *dan tien* and considered the central engine of internal energy in the body in Chinese martial arts.[5](#)

Stanford University physicist William Tiller constructed an ingenious device to measure the energy produced by healers. The equipment discharged a steady stream of gas and recorded the exact number of electrons pulsing out with the discharge. Any increase in voltage would be captured by the pulse counter.

In his experiment, Tiller asked ordinary volunteers to place their hands about six inches from his device and hold a mental intention to increase the count rate. In the majority of more than 1000 such experiments, Tiller discovered that, during the intention, the number of recorded pulses would increase by 50,000 and remain there for 5 minutes.

These increases would occur even if a participant was not close to the machine, so long as he or she held an intention.

Tiller concluded that directed thoughts produce demonstrable physical energy, even over remote distance.[6](#)

Tiller concluded that directed thoughts produce demonstrable physical energy, even over remote distance..

I found two other studies measuring the actual electrical frequencies emitted by people using intention.

One study measured healing energy and the other examined energy generated by a Chinese *Qigong* master during times that he was emitting external *Qi*, the Chinese term for energy or the life force.[7](#)

In both instances, the measurements were identical: frequency levels of 2–30 hertz were being emitted by the healers.

This energy also seemed to change the molecular nature of matter.

Qi seemed to change the molecular nature of matter.

I discovered a body of scientific evidence examining chemical changes caused by intention.

Bernard Grad, an associate professor of biology at McGill University in Montreal had examined the effect of healing energy on water that was to be used to irrigate plants. After a group of healers had sent healing to samples of water, Grad chemically analysed the water by infrared spectroscopy.

He discovered that the water treated by the healers had undergone a fundamental change in the bonding of oxygen and hydrogen in its molecular makeup.

The hydrogen bonding between the molecules had lessened in a similar manner to that which occurs in water exposed to magnets.[8](#)

A number of other scientists confirmed Grad's findings; Russian research discovered that the hydrogen–oxygen bonds in water molecules undergo distortions in the crystalline microstructure during healing.[9](#)

These kinds of changes can occur simply through the act of intention.

In one study, experienced meditators sent an intention to affect the molecular structure of water samples they were holding throughout the meditation. When the water was later examined by infrared spectrophotometry, many of its essential qualities, particularly its absorbance – the amount of light absorbed by the water at a particular wavelength had been significantly altered.[10](#)

When someone holds a focused thought, he may be altering the very molecular structure of the object of his intention.

When someone holds a focused thought, he may be altering the very molecular structure of the object of his intention.

In his research, Gary Schwartz wondered whether intention only manifested as electrostatic energy. Perhaps magnetic energy also played a role.

Magnetic fields naturally had more power, more 'push–pull' energy. Magnetism seemed the more powerful and universal energy; the earth itself is profoundly influenced by its own faint pulse of geomagnetic energy.

Schwartz remembered a study carried out by William Tiller, in which psychics had been placed inside a variety of devices that block different forms of energy. They had performed better than usual in a Faraday cage, which filters out only electrical energy, but they performed worse when placed in a magnetically shielded room.[11](#)

From these early studies, Schwartz gleaned two important implications: healing may generate an initial surge of electricity, but the real transfer mechanism may be magnetic.

Indeed, psychic phenomena and psychokinesis could be differentially influenced, simply through different types of shielding. Electrical signals might interfere, while magnetic signals enhance the process.

To test this latest idea, Schwartz was approached by a colleague of his, Melinda Connor, a post-doctoral fellow in her mid-forties with an interest in healing.

The first hurdle was finding an accurate means of picking up magnetic signals. Measuring tiny low-frequency magnetic fields is tricky, requiring the use of expensive and highly sensitive equipment called a SQUID, or superconducting quantum interference device. A SQUID, which can cost up to four million dollars, ordinarily occupies a specially constructed room that has been magnetically shielded in order to eliminate ambient radiating noise.

The best Schwartz and Connor could come up with on their limited budget was a poor man's SQUID – a small handheld, battery-operated three-axis digital gaussmeter originally designed to measure electromagnetic pollution by picking up extra-low-frequency (ELF) magnetic fields.

The gaussmeter was sensitive enough to pick up one-thousandth of a gauss, a very faint pulse of a magnetic field. In Schwartz's mind, this level of sensitivity was more than adequate to do the job.

It occurred to Connor that the way to measure change in low-frequency magnetic fields was to count the number of changes in the meter reading over time. When simply recording ambient stable magnetic fields, the device will only deviate slightly by less than one-tenth of a gauss.

However, in the presence of an oscillating magnetic field – with periodic changes in frequency – the numbers will keep moving, from, say, 0.6 to 0.7 to 0.8, and back down to 0.6.

The greater and more frequent the change, which would be recorded by the number of changes in the dials, the more likely it is that the magnetic field has been affected by a source of directed energy.

Connor and Schwartz gathered together a group of practitioners of Reiki, the healing art developed a century ago in Japan.

They took measurements near each hand of all the healers during alternating periods while they were ‘running energy’ and then during times they were at rest, with their eyes closed. Next, the pair assembled a group of ‘master healers’ with a substantial track record of successful, dramatic healings. Again, Connor and Schwartz took magnetic field measurements near each hand, while the master healers were running energy and at rest. Then, they compared the Reiki measurements with measurements they had taken of people who had not been trained in healing.

Once Schwartz and Conner had analyzed the data, they discovered that both groups of healers demonstrated significant fluctuations in very low pulsations of a magnetic field, emanating from both hands.

A huge increase in oscillations in the magnetic field occurred whenever a healer began to run energy. However, the most profound energy increase surged from their dominant hands. The control group of people who were not trained healers did not demonstrate the same effect.

Then Schwartz compared effects from the Reiki group with those of the master healers and discovered another enormous difference. The master healers averaged close to a third more magnetic-field changes per minute than the Reiki healers.[12](#)

The study results seemed clear. Schwartz and Connor had their proof that directed intention manifests as both electrostatic *and* magnetic energy.

Directed intention manifests as both electrostatic and magnetic energy. But they also discovered that intention was like playing the piano; you need to learn how to do it, and some people do it better than others.

The ability to manipulate energy comes with training and practice.

In considering what this all meant, Gary Schwartz thought of the phrase often used by medical doctors, usually in emergency situations: *when you hear hoof beats, don't think zebras* .

In other words, when you are trying to diagnose someone with physical symptoms, first rule out all the most likely causes, and only then consider more exotic possibilities.

He liked to approach science in the same way and so he questioned his own findings: Could the healers' increase in magnetic field oscillations during healing simply be the result of certain peripheral biophysical changes? Muscle contractions generate a magnetic field, as do changes in blood flow, the increasing or decreasing dilation of blood vessels, the body's current volume of liquid or even the flow of electrolytes. Skin, sweat glands, change of temperature, neural induction – all generate magnetic fields.

His guess was that healing resulted from a summation of multiple biological processes that are mediated magnetically.

But the possibility that healing might be a magnetic effect did not explain long-distance remote healing.

In some instances, healers sent healing from thousands of miles away and the effect did not decay with distance. In one successful study of AIDS patients who improved through remote healing, the 40 healers involved in the study sent the healing to the San Francisco patients from locations all across America.[13](#)

Similar to electrical fields, magnetic fields decrease with distance. The magnetic and electrical effects were likely to be some aspect of the process, but not its central one. It was likely to be closer to a quantum field, possibly more akin to light.

Schwartz began to consider the possibility that the mechanism creating intention originated with the tiny elements of light emitted from human beings. In the mid-1970s, a German physicist named Fritz-Albert Popp had stumbled upon the fact that all living things, from the most basic of single-celled plants to the most sophisticated of organisms like human beings, emitted a constant tiny current of photons – tiny particles of light.[14](#)

He labelled them 'biophoton emissions' and believed that he had uncovered the primary communication channel of a living organism – that it used light as a means of signalling to itself and to the outside world.

For more than 30 years, Popp has maintained that this faint radiation, rather than biochemistry, is the true driving force in orchestrating and coordinating all cellular processes in the body. Light waves offered a perfect communication system able to transfer information almost instantaneously across an organism. Having waves, rather than chemicals, as the communication mechanism of a living being also solved

the central problem of genetics – how we grow and take final shape from a single cell. It also explains how our bodies manage to carry out tasks with different body parts simultaneously. Popp theorized that this light must be like a master tuning fork setting off certain frequencies that would be followed by other molecules of the body.[15](#)

A number of biologists, such as the German biophysicist Herbert Fröhlich, had proposed that a type of collective vibration causes proteins and cells to coordinate their activities.

Nevertheless, all such theories were ignored until Popp's discoveries, largely because no equipment was sensitive enough to prove they were right.

With the help of one of his students, Popp constructed the first such machine – a photomultiplier that captured light and counted it, photon by photon. He carried out years of impeccable experimentation that demonstrated that these tiny frequencies were mainly stored and emitted from the DNA of cells.

The intensity of the light in organisms was stable, ranging from a few to several hundred photons per second per square centimetre surface of the living thing – until the organism was somehow disturbed or ill, at which point the current went sharply up or down.

The signals contained valuable information about the state of the body's health and the effects of any particular therapy. Cancer victims had fewer photons, for instance. It was almost as though their light were going out.

Initially vilified for his theory, Popp was eventually recognized by the German government and then internationally.

Eventually he formed the International Institute of Biophysics (IIB), composed of 15 groups of scientists from international centres all around the world, including prestigious institutions like CERN in Switzerland Northeastern University in the USA, the Institute of Biophysics Academy of Scienc in Beijing, China, and Moscow State University in Russia. By the early twenty-firs century, the IIB numbered at least 40 distinguished scientists from around the globe.

Could it be that these were the frequencies that mediated healing? Schwartz realized that if he was going to carry out studies of biophoton emissions, first he had to figure out how to view these tiny emissions of light.

In his laboratory, Popp developed a computerized mechanism attached to a box in which a living thing, such as a plant, could be placed. The machine could count the photons and chart the amount of light emitted on a graph. But those machines only

recorded photons in utter pitch blackness. Up until then, it had been impossible for scientists to witness living things actually glowing in the dark.

As Schwartz mulled over the kind of equipment that would allow him to see very faint light, he thought of state-of-the-art supercooled charge-coupled device (CCD) cameras on telescopes. This exquisitely sensitive equipment, now used to photograph galaxies deep in space, picks up about 70 per cent of any light, no matter how faint.

CCD devices were also used for night-vision equipment.

If a CCD camera could pick up the light from the most distant of stars, it might also be able to pick up the faint light coming off living things. However, this kind of equipment can cost hundreds of thousands of dollars and usually had to be cooled to temperatures only 100 degrees above absolute zero, to eliminate any ambient radiation emitted at room temperature. Cooling the camera down also helped to improve its sensitivity to faint light. Where on earth was he going to get hold of this kind of high-tech equipment?

Kathy Creath, a professor of optical sciences at Schwartz's university, who shared his fascination with living light and its possible role in healing, had an idea. As it happened, she knew that the department of radiology at the National Science Foundation (NSF) in Tucson owned a low-light CCD camera, which they used to measure the light emitted from laboratory rats after being injected with phosphorescent dyes.

The Roper Scientific VersArray 1300 B low-noise, high performance CCD camera was housed in a dark room inside a black box and above a Cryotiger cooling system, which cooled temperatures to -100°C . A computer screen displayed its images. It was just what they were looking for. After Creath approached the director of the NSF project, he generously agreed to allow the two of them access to the camera during its down time.

In their first test, Schwartz and Creath placed a geranium leaf on a black platform. They took fluorescent photographs after exposures of up to five hours. When the computer displayed the final photograph, it was dazzling: a perfect image of the leaf in light, like a shadow in reverse, but in incredible detail, each of its tiniest veins delineated.

Surrounding the leaf were little white spots, like a sprinkling of fairy dust – evidence of high-energy cosmic rays. With his next exposure, Schwartz used a software filter to screen out the ambient radiation. The image of the leaf was now perfect.

As they studied this latest photograph on the screen of the computer in front of them, Schwartz and Creath understood that they were making history. It was the first time a scientist had been able to witness images of the light actually emanating from a living thing.[16](#)

Now that he had equipment that captured and recorded light, Schwartz was finally able to test whether healing intention also generated light.

Creath got hold of a number of healers, and asked them to place their hands on the platform underneath the camera for 10 minutes. Schwartz's first crude images showed a rough glow of large pixilations, but they were too out of focus for him to analyse them.

Next he tried placing the healers' hands on a white background (which reflected light) rather than on a black background (which absorbed light). The images were breathtakingly clear: a stream of light flowed out of the healers' dominant hands, almost as though it were flowing from their fingers. Schwartz now had his answer about the nature of conscious thought: healing intention creates waves of light – and, indeed, among the most organized light waves found in nature.

The theory of relativity was not Einstein's only great insight.

He had had another astonishing realization in 1924, after correspondence with an obscure Indian physicist, Satyendra Nath Bose, who had been pondering the then-new idea that light was composed of little vibrating packets called photons. Bose had worked out that, at certain points, photons should be treated as identical particles. At the time nobody believed him – nobody but Einstein, after Bose sent him his calculations.

Einstein liked Bose's proofs and used his influence to get Bose's theory published. Einstein also was inspired to explore whether, under certain conditions or certain temperatures, atoms in a gas, which ordinarily vibrated anarchically, might also begin to behave in synchrony, like Bose's photons. Einstein set to work on his own formula to determine which conditions might create such a phenomenon.

When he reviewed his figures, he thought he had made a mistake in his calculations.

According to his results, at certain extraordinarily low temperatures, just a few kelvin above absolute zero, something really strange would begin to happen: the atoms, which ordinarily can operate at a number of different speeds, would slow down to identical energy levels. In this state, the atoms would lose their individuality and both look and behave like one giant atom. Nothing in his mathematical armamentarium could tell them apart. If this were true, he realized, he had stumbled upon an entirely

new state of matter, with utterly different properties from anything known in the universe.

Einstein published his findings,[17](#) and lent his name to the phenomenon, called a Bose–Einstein condensate, but he was never convinced that he had been right.

Nor were other physicists, until more than 70 years later when, on 5 June 1995, Eric Cornell and Carl Wieman of JILA, a programme sponsored by the National Institute of Standards and Technology and the University of Colorado at Boulder, managed to cool a tiny batch of rubidium atoms down to 170 billionths of a degree above absolute zero.[18](#)

It had been quite a feat, requiring trapping the atoms in a web of laser light and then magnetic fields. At a certain point, a batch of some 2000 atoms – measuring about 20 microns, about one-fifth the thickness of a single piece of paper – began behaving differently from the cloud of atoms surrounding them, like one smeared-out single entity. Although the atoms were still part of a gas, they were behaving more like the atoms of a solid.

Four months later, Wolfgang Ketterle from Massachusetts Institute of Technology replicated their experiment, but with a form of sodium, for which he, as well as Cornell and Wieman, won the 2001 Nobel prize.[19](#)

Then a few years after that, Ketterle and others like him were able to reproduce the effect with molecules.[20](#)

Scientists believed that a form of Einstein and Bose’s theory could account for some of the strange properties they had begun to observe in the subatomic world: superfluidity, when certain fluids can flow without losing energy, or even spontaneously work themselves out of their containers; or superconduction, a similar property of electrons in a circuit. In superfluid or superconductor states, liquid or electricity could theoretically flow at the same pace forever.

Ketterle had discovered another amazing property of atoms or molecules in this state. All the atoms were oscillating in perfect harmony, similar to photons in a laser, which behave like one giant photon, vibrating in perfect rhythm. This organization makes for an extraordinary efficiency of energy. Instead of sending a light about 3 meters, the laser emits a wave 300 million times that far.

Scientists were convinced that a Bose–Einstein condensate was a peculiar property of atoms and molecules slowing down so much that they are almost at rest, when exposed to temperatures only a fraction above the coldest temperatures in the universe.

But then Fritz-Albert Popp and the scientists working with him made the astonishing discovery that a similar property existed in the weak light emanating from organisms. This was not supposed to happen in the boiling inner world of the living thing. What is more, the biophotons he measured from plants, animals and humans were highly coherent. They acted like a single super-powerful frequency, a phenomenon also referred to as 'superradiance'.

The German biophysicist Herbert Fröhlich had first described a model in which this type of order could be present and play a central role in biological systems. His model showed that, with complex dynamic systems like human beings, the energy within created all sorts of subtle relationships, so that it is no longer discordant.[21](#)

Living energy is able to organize to one giant coherent state, with the highest form of quantum order known to nature.

When subatomic particles are said to be 'coherent', or 'ordered', they become highly interlinked by bands of common electromagnetic fields, and resonate like a multitude of tuning forks all attuned to the same frequency. They stop behaving like anarchic individuals and begin operating like one well-rehearsed marching band.

As one scientist put it, coherence is like comparing the photons of a single 60-watt light bulb to the sun.

Ordinarily, light is extraordinarily inefficient. The intensity of light from a bulb is only about 1 watt per square centimetre of light – because many of the waves made by the photons destructively interfere with and cancel out each other. The light per square centimetre generated by the sun is about 6000 times stronger. But if you could get all the photons of this one small light bulb to become coherent and resonate in harmony with each other, the energy density of the single light bulb would be thousands to millions of times higher than that of the surface of the sun.[22](#)

After Popp made his discoveries about coherent light in living organisms, other scientists postulated that mental processes also create Bose–Einstein condensates. British physicist Roger Penrose and his partner, American anaesthetist Stuart Hameroff from the University of Arizona, were in the vanguard of frontier scientists who proposed that the microtubules in cells, which create the basic structure of the cells, were 'light pipes' through which disordered wave signals were transformed into highly coherent photons and pulsed through the rest of the body.[23](#)

Gary Schwartz had witnessed just this coherent photon stream emanating from the hands of healers. After studying the work of scientists like Popp and Hameroff, he finally had his answer about the source of healing: if thoughts are generated as frequencies, healing intention is well-ordered light.

Gary Schwartz's creative experiments revealed to me something fundamental about the quantum nature of thoughts and intentions. He and his colleagues had uncovered evidence that human beings are both receivers and transmitters of quantum signals. Directed intention appears to manifest as both electrical and magnetic energy and to produce an ordered stream of photons, visible and measurable by sensitive equipment. Perhaps our intentions also operate as highly coherent frequencies, changing the very molecular makeup and bonding of matter. Like any other form of coherence in the subatomic world, one well-directed thought might be like a laser light, illuminating without ever losing its power.

I was reminded of an extraordinary experience Schwartz once had in Vancouver. He had been staying in the penthouse apartment suite of a downtown hotel. He had awakened at 2 a.m., as he often did, and had walked out to the balcony to have a look at the spectacular view of the city to the west, framed by the mountains. He was surprised to see how many hundreds of homes along the peninsula below him still had their lights on.

He wished he had a telescope handy to see what some of the people were doing up at this late hour. But of course, if any of them had their own telescope, they would be able to see him standing there in the nude. An odd thought suddenly came to him of his own naked image flying into each window. But maybe the idea was not so fanciful.

After all, he was emitting a constant stream of biophotons, all travelling at the speed of light; each photon would have travelled 186,000 miles one second later, and 372,000 miles one second after that.

His light was not unlike the photons of visible light emanating from stars in the sky. Much of the light from distant stars has been traveling for millions of years. Starlight contains a star's individual history. Even if a star had died long before its light reached earth, its information remains, an indelible footprint in the sky.

He then had a sudden image of himself as a ball of energy fields, a little star, glowing with a steady stream of every photon his body had ever produced for more than 50 years.

All the information he had been sending from the time he was a young boy in Long Island, every last thought he had ever had, was still out there, glowing like starlight. Perhaps, I thought, intention was also like a star. Once constructed, a thought radiated out like starlight, affecting everything in its path.

Notes – Chapter 2: The Human Antenna

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